



FFF Printing Fundamentals

Part 1: Process

3D Printing is a Process

1) Start with a 3D design.

- Check the license for restrictions when you download.
- *STL*. Approximates shapes with triangles.
- *STEP*. Real geometry. Great for 3D modeling.
- *3MF*. Compressed container, like a zip file.
Multiple objects, slicer settings, multi-color settings, ...

2) Slice and process the design with software.

- Printers do not accept 3D models!
- “Slice” objects into vertically-stacked layers.
- Create G-code file. List of toolpaths and commands that encode:
Coordinates, speeds, temperatures, ...

3D Printing is a Process

3) Print.

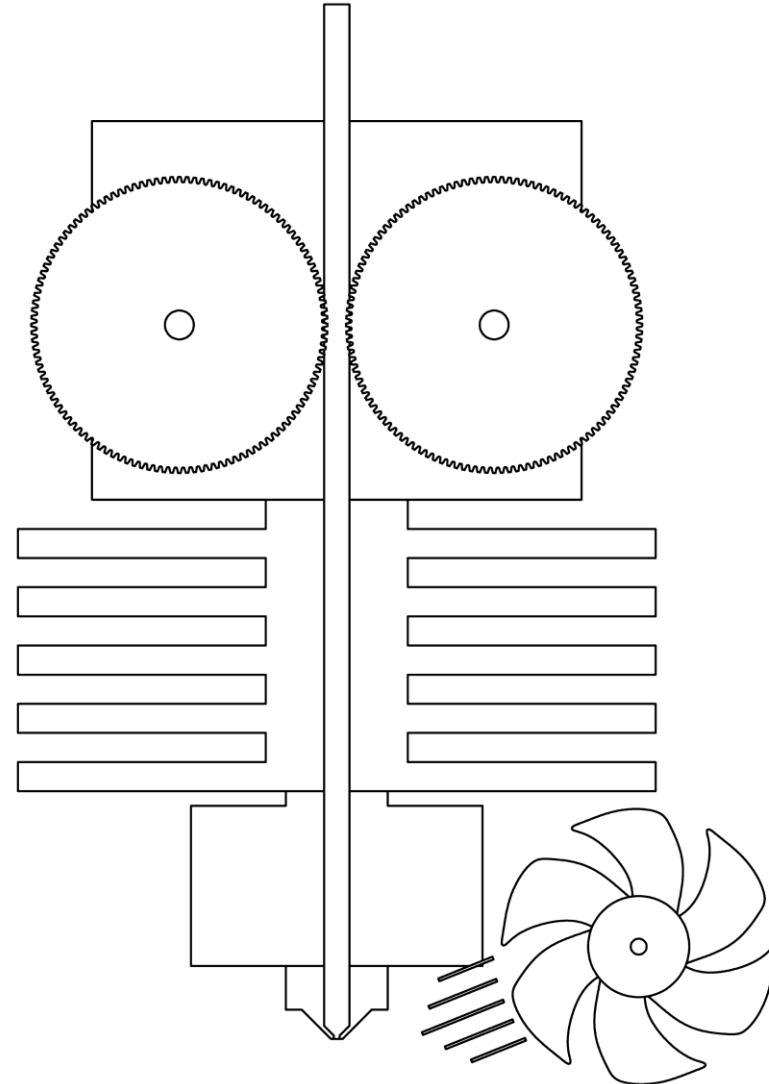
- Do the pieces fit?
- Does it work?
- Do you need to change the design? Print settings?

4) Post-process.

- Remove supports.
- Assemble.
- Glue.
- Sand or file.
- Paint.

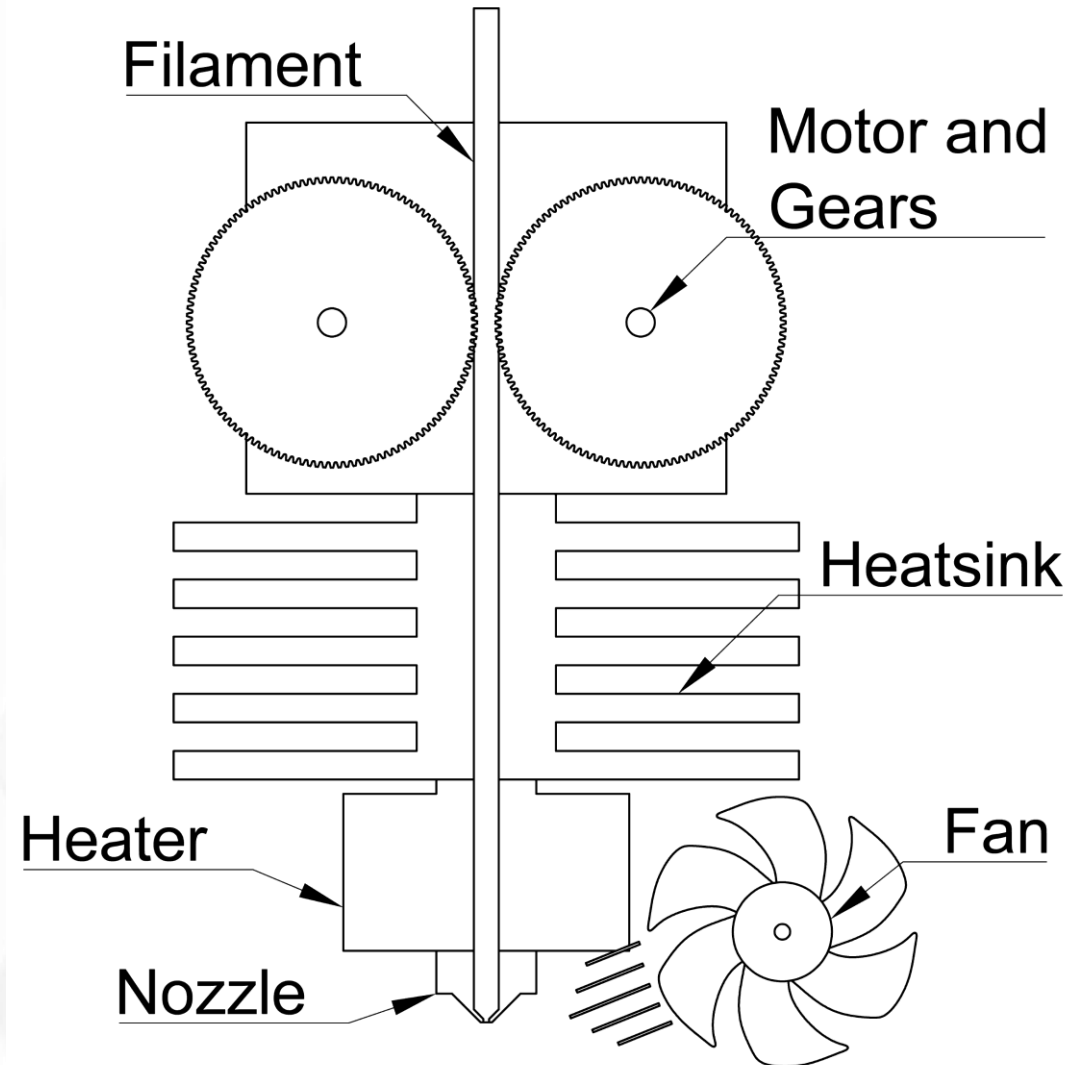
Pushing Plastic

- What kind of 3D printing?
- Filament and extrusion-based additive manufacturing.
- **FFF. Fused filament fabrication.**
- FDM™. Fused deposition modeling.
 - Essentially the same but owned by Stratasys.
- **Toolhead.** Moves around and deposits plastic.



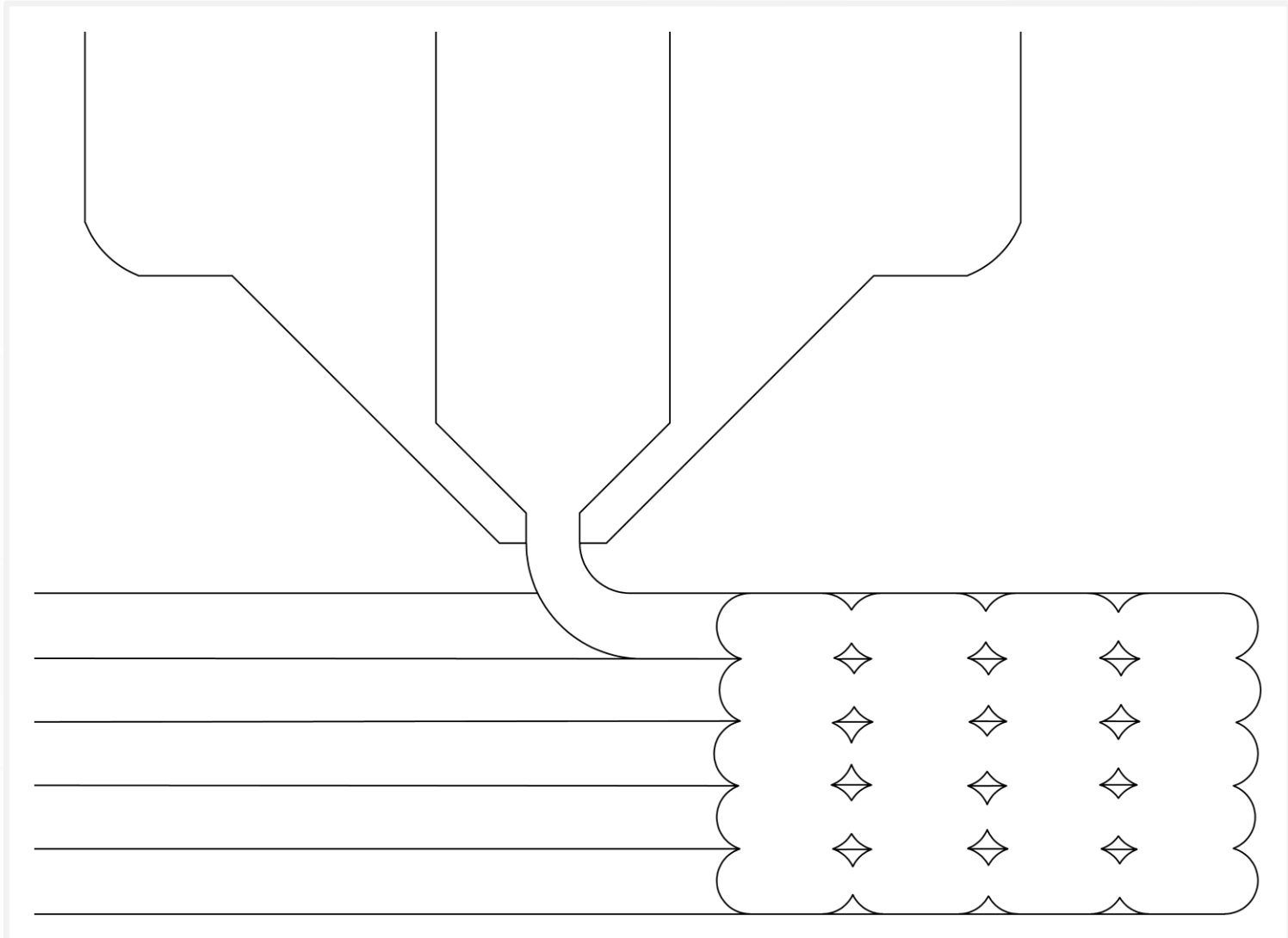
Pushing Plastic

- **Filament.** Long, continuous plastic rod.
- **Motor and gears** grab and push filament.
- **Heatsink** keeps heat in the hotend (heater + nozzle).
- **Heater** softens filament at the glass transition temperature.
- **Nozzle** opening shapes plastic.
- **Fan** cools hot plastic.



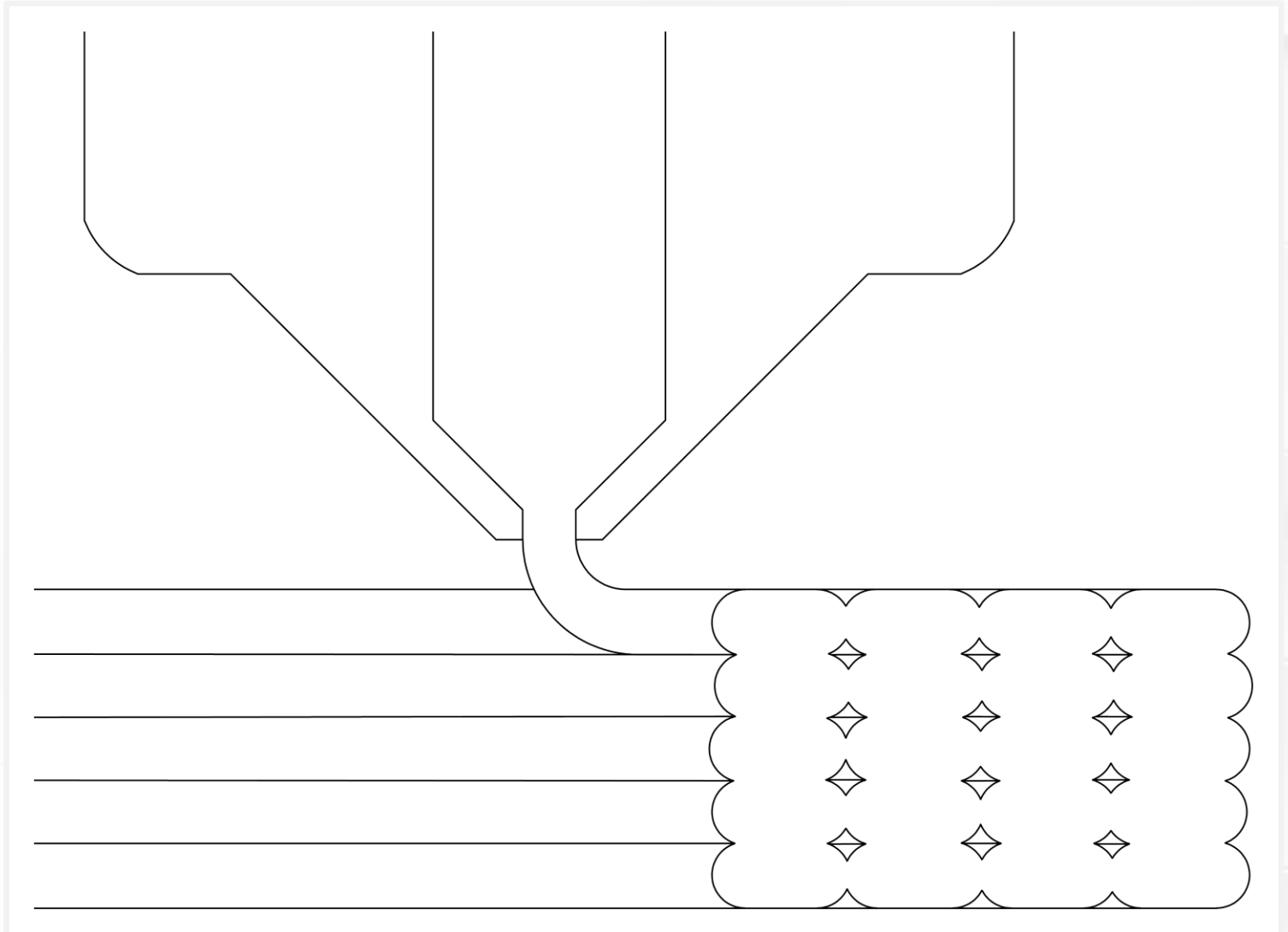
Extrusions

- Toolhead deposits lines called **extrusions**.
- Best to squish extrusions into place.
- Lock in place as they cool.



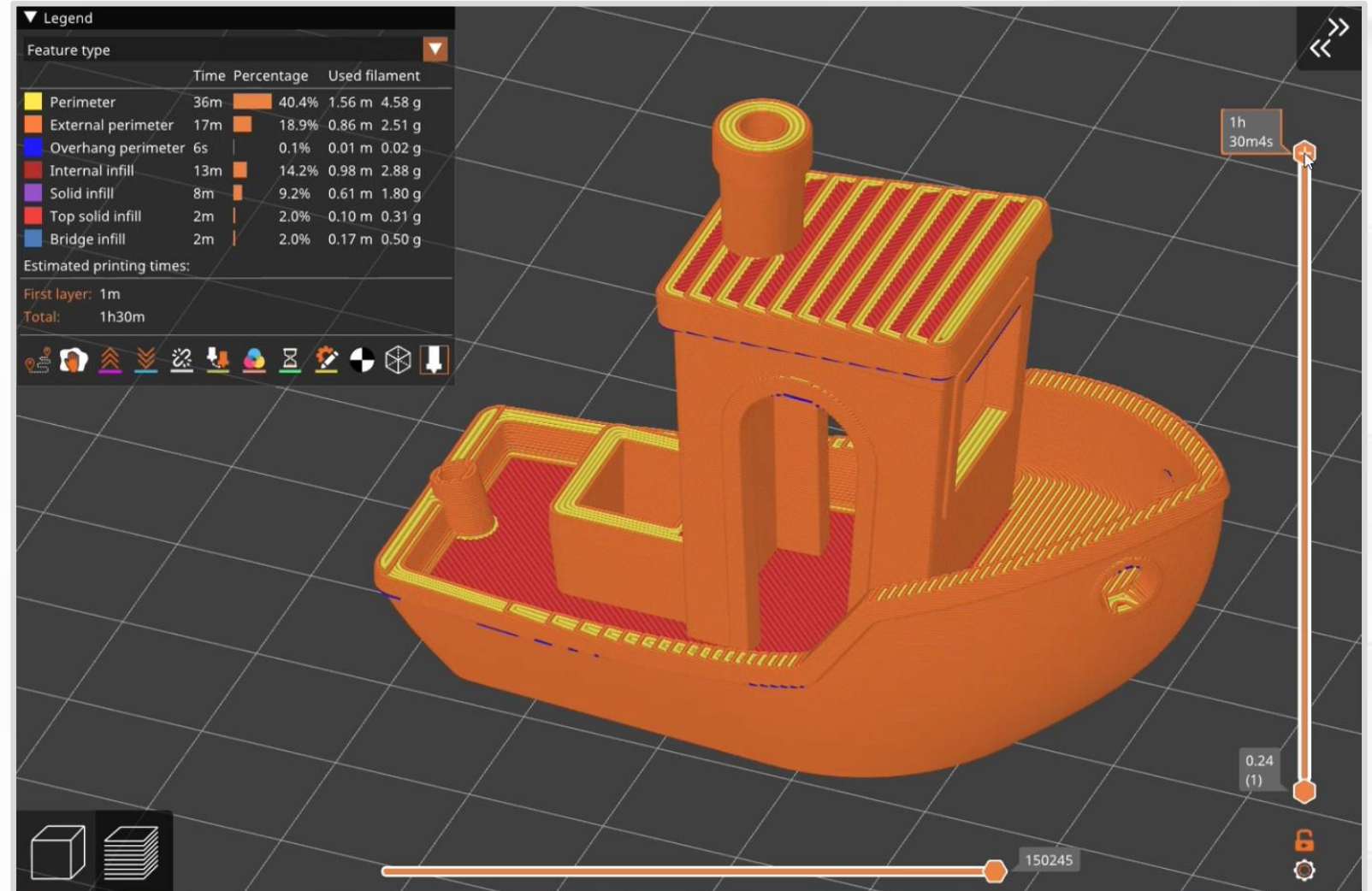
Stacking Layers

- Printers simulate 3D shapes by stacking 2D layers.
- You could average the two and call it 2.5D printing.



Slicer Layers

Let's see how PrusaSlicer handles layers.



(Animation,
[FlyingGyroscope GitHub](#))

Slicer Layers

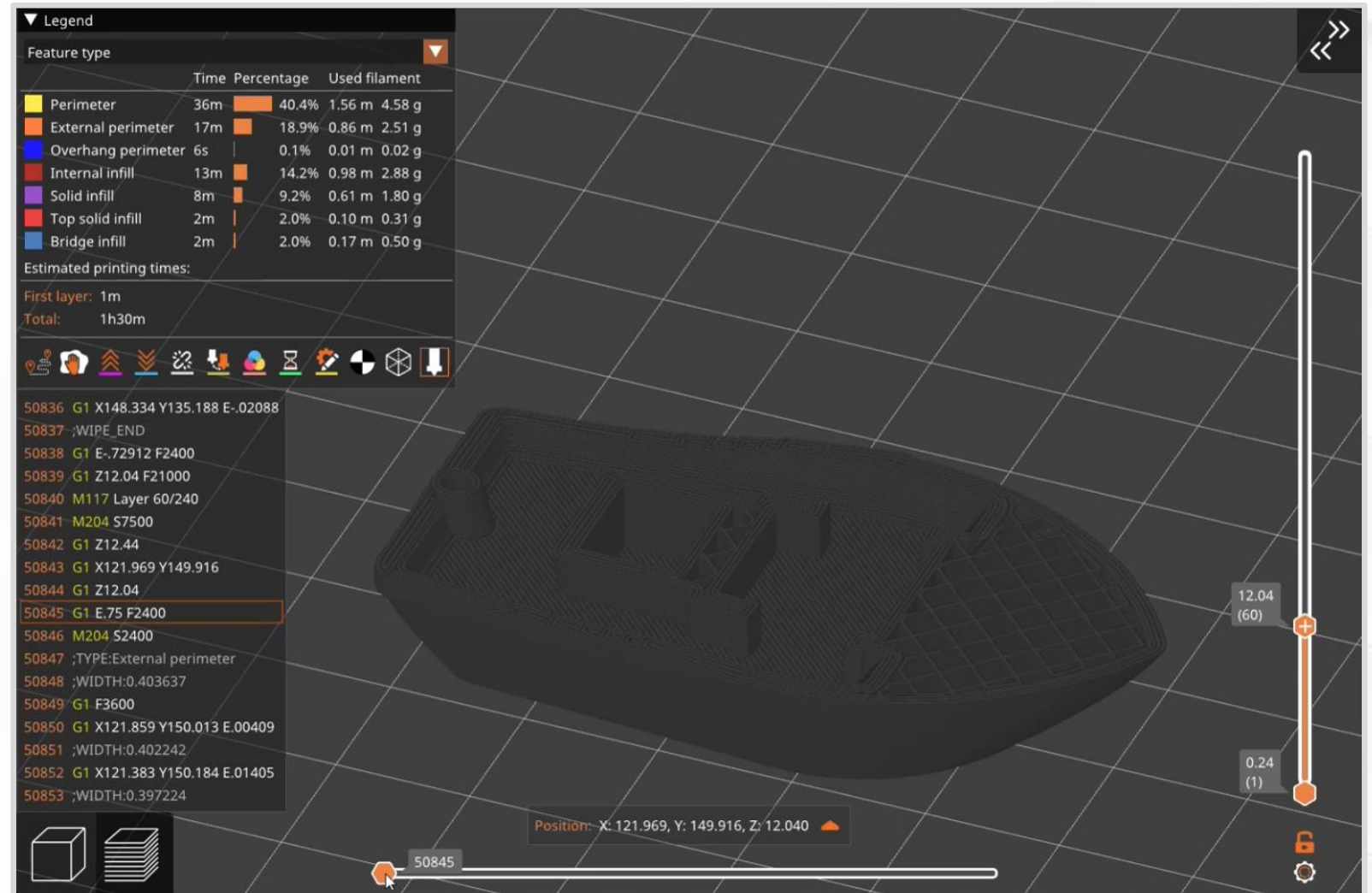
- Sort extrusions by shape and geometry.
- Color-coded.
- Adjust settings for each category. (Mostly printing speed.)

▼ Legend					
Feature type ▼					
		Time	Percentage	Used filament	
	Perimeter	36m		40.4%	1.56 m 4.58 g
	External perimeter	17m		18.9%	0.86 m 2.51 g
	Overhang perimeter	6s		0.1%	0.01 m 0.02 g
	Internal infill	13m		14.2%	0.98 m 2.88 g
	Solid infill	8m		9.2%	0.61 m 1.80 g
	Top solid infill	2m		2.0%	0.10 m 0.31 g
	Bridge infill	2m		2.0%	0.17 m 0.50 g
Estimated printing times:					
First layer: 1m					
Total: 1h30m					

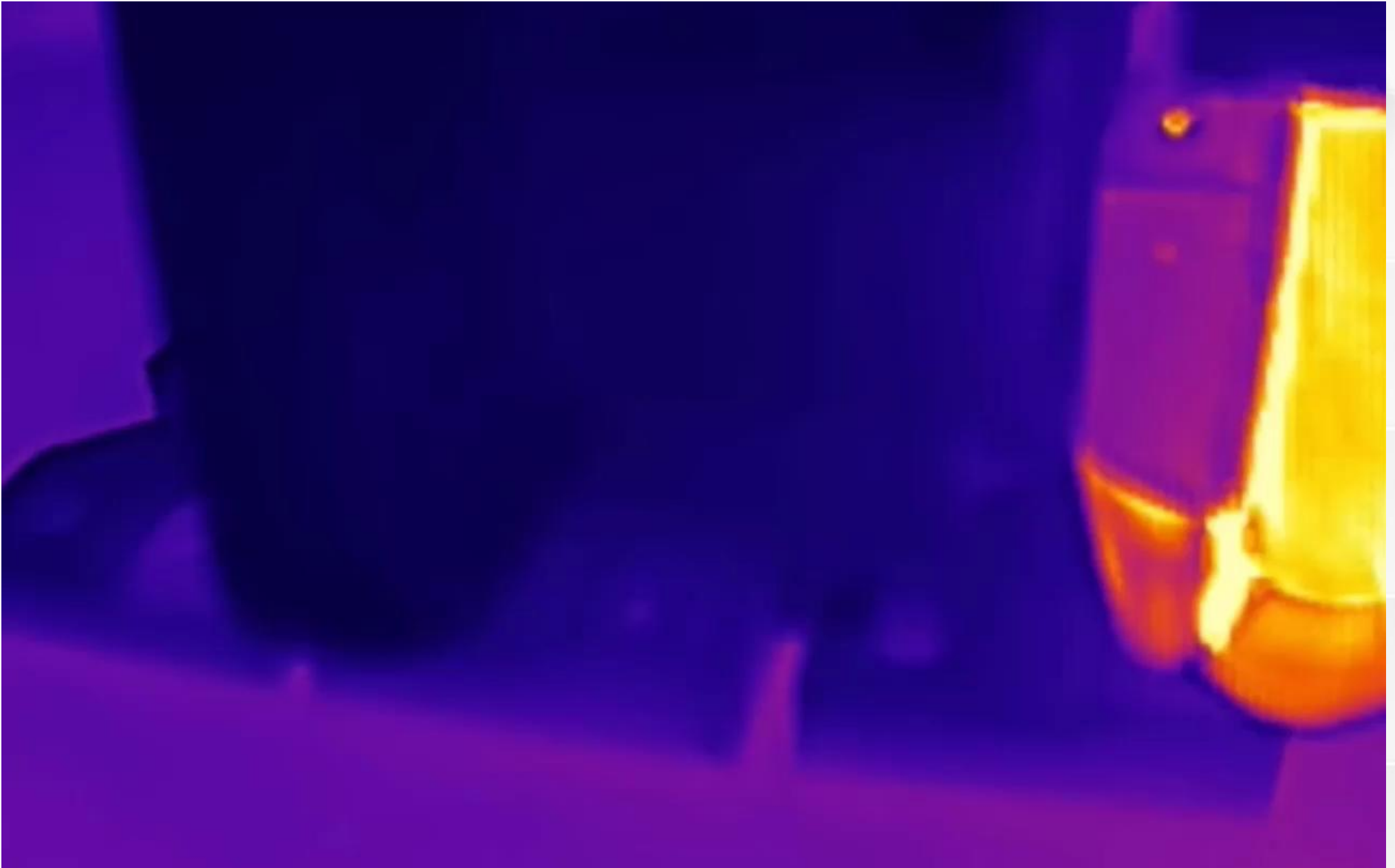
Slicer Layers

- Single layer preview.
- **G-code** shows what the printer is actually doing (lower left).
- *G1* = Move. Include X, Y, or Z coordinates.
- *G1 X Y E* = Move and extrude.

(Animation,
FlyingGyroscope GitHub)



(Animation,
Thermal camera
video by
NeedItMakelt)



Coordinating Settings

- Making consistent extrusions requires tuned settings and special techniques.
- **Hotend temperature.** Determines how easily filament flows.
 - Changes viscosity and affects all flow-related settings.
- **Part cooling fan.** Dynamically adjust cooling to hold extrusions in place.
Extra cooling for
 - Quick layers, based on minimum layer time.
 - Sagging extrusions on overhangs and bridges.
- **Extrusion Multiplier (EM, EX).** Fine tune the overall flow rate.
 - Too low = under-extruding. Gaps between lines, weak parts.
 - Too high = over-extruding. Excess material builds up, rough surfaces, offset dimensions.

Coordinating Settings

- **Maximum volumetric flow.** Stay within physical limits of hotend.
 - Prevent under-extruding at high speeds.
- **Pressure/Linear advance.** Compensate for uneven flow during changes.
 - *Example:* Pressure gradually builds at the start of an extrusion, so momentarily overshoot at the beginning.
- **Retraction.** Pull back filament and the end of an extrusion.
 - Stop flow while toolhead moves to a new location.
 - Prevent leaks, oozing, stringing, blobs.
- Check out community guides for more information.
 - [Ellis' Print Tuning Guide](#).
 - [OrcaSlicer Calibration Guide](#).

Easy Filaments

- **PLA** (derived from corn starch).
 - General-purpose filament for indoor applications.
 - Excellent bridging.
 - Very stiff.
 - Creeps. Deforms permanently under load.
 - Low temperature resistance. Softens inside a car on a hot summer day.
- **PETG** (plastic water bottles).
 - Pretty good general-purpose filament.
 - Hygroscopic. Needs low-humidity storage.
 - Elastic. Excellent choice for springs.
 - Moderate temperature resistance. Can be used outdoors.

Easy Filaments

- What makes PLA and PETG easy?
 - Low warp. Low shrink.
 - High dimensional accuracy.
 - Room temperature, open-air (non-enclosed) print chamber.

Proper Protocol

Let's review some everyday guidelines.

- Secure the loose end of a spool of filament at all times.
 - Tangles ruin prints and waste time.
- Do not reach into the printer while it is running.
 - Use touchscreen or buttons to pause and resume prints.
- Watch the first layer before you walk away.
- Avoid burns.
 - Read nozzle and print bed temperatures on the display.
 - Never reach near a hot nozzle (210 °C = 410 °F).
- Avoid leaving fingerprints on print sheets.
 - Clean sheets as they get dirty.